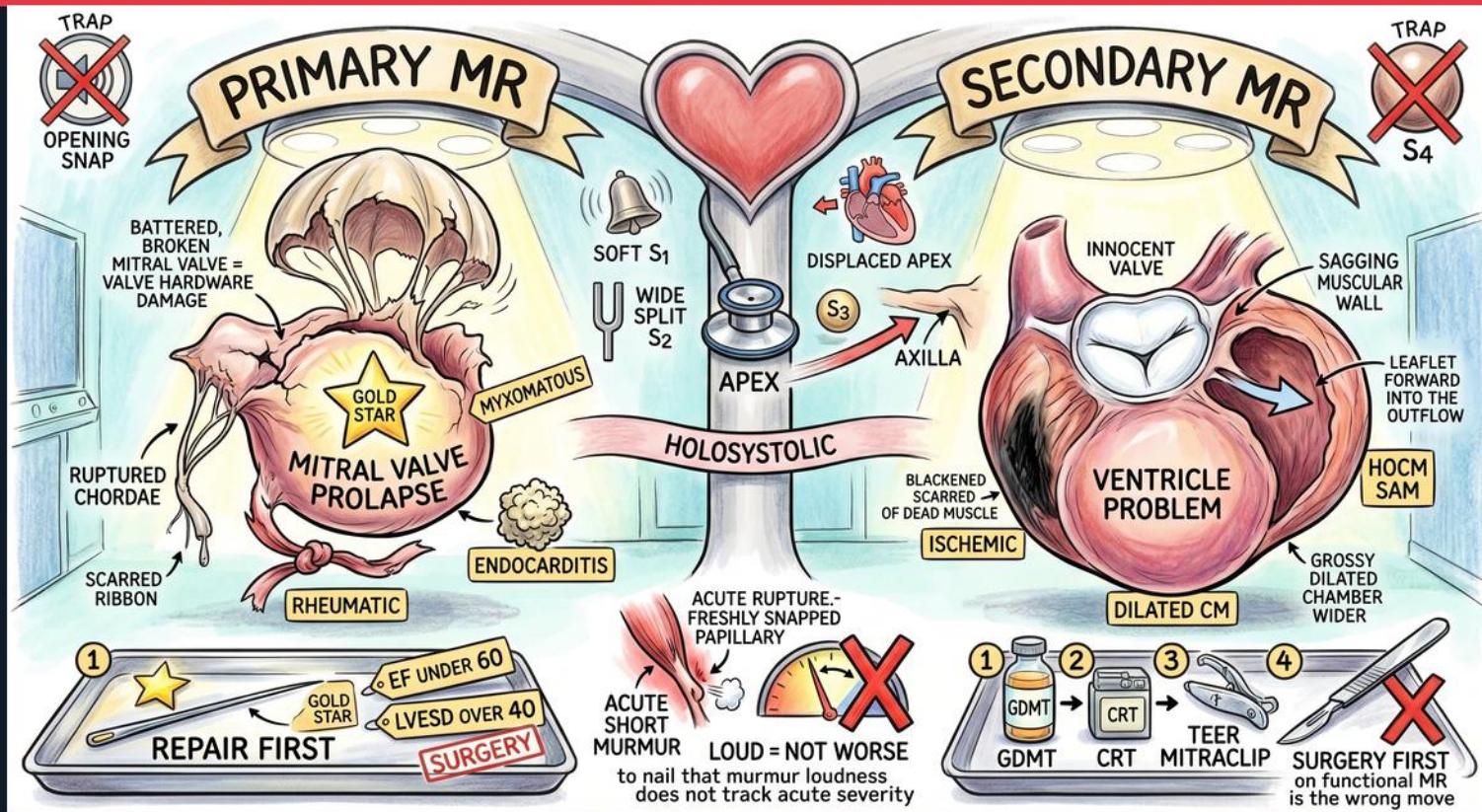
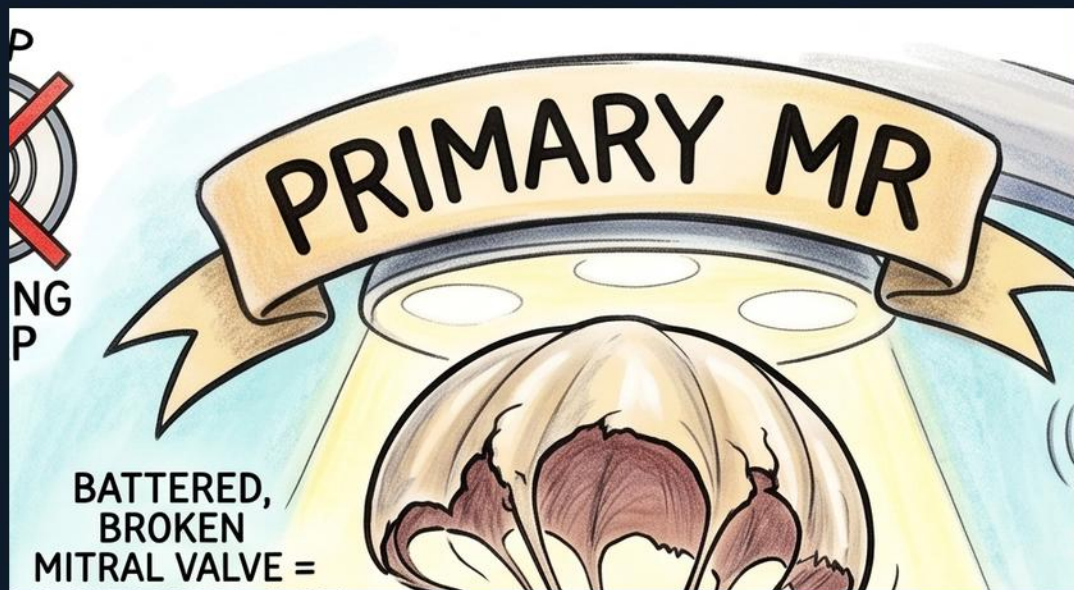


Valvular — Mitral Regurgitation (Primary vs Secondary)





Primary MR (valve problem)

WHAT YOU SEE

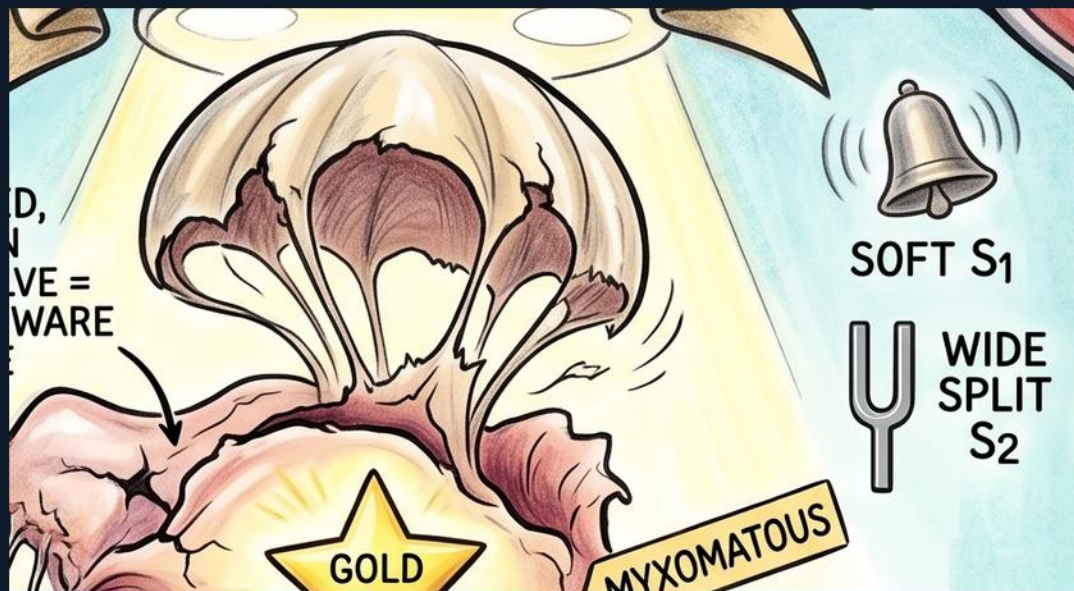
Left banner reading PRIMARY MR over a parachute-like mitral valve apparatus.

WHAT IT ENCODES

Primary (organic) MR means the valve leaflets/apparatus themselves are diseased while the ventricle is initially normal. The leak is driven by structural valve damage, so the definitive fix is mechanical — repair or replace the valve.

THE BOARD TRAP

Treating any severe MR as primary — first decide valve problem (primary) vs ventricle problem (secondary); management diverges.



Mitral valve prolapse / myxomatous

WHAT YOU SEE

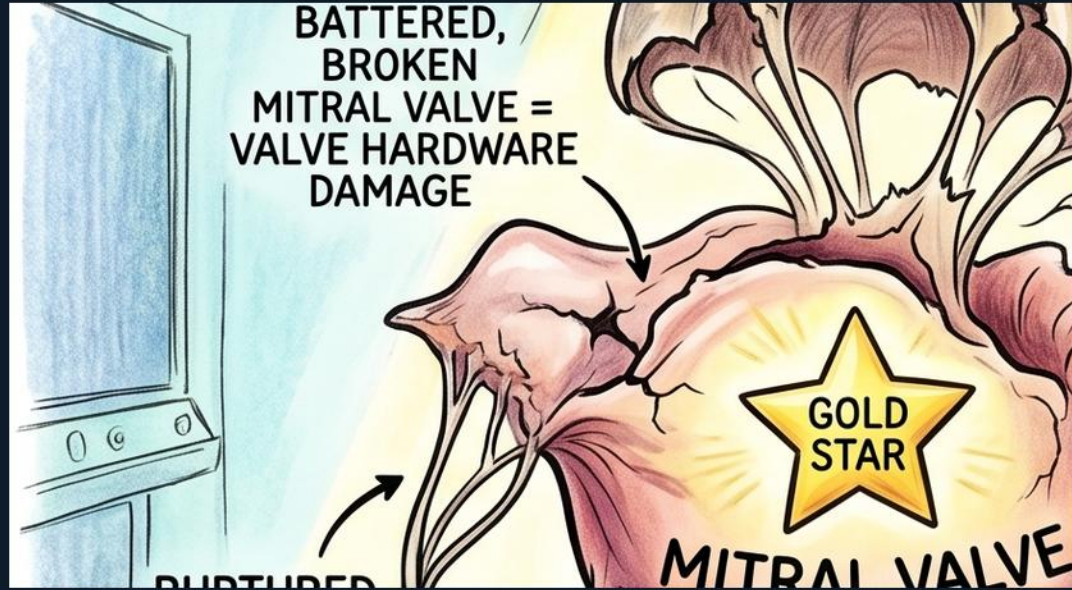
The billowing parachute labeled MITRAL VALVE PROLAPSE with a GOLD STAR and tag MYXOMATOUS.

WHAT IT ENCODES

Myxomatous degeneration causing mitral valve prolapse is the single most common cause of primary MR in developed countries. Redundant, billowing leaflets fail to coapt and prolapse into the LA during systole.

THE BOARD TRAP

Naming rheumatic disease as the most common cause of primary MR — in developed countries it is myxomatous MVP.



Ruptured chordae

WHAT YOU SEE

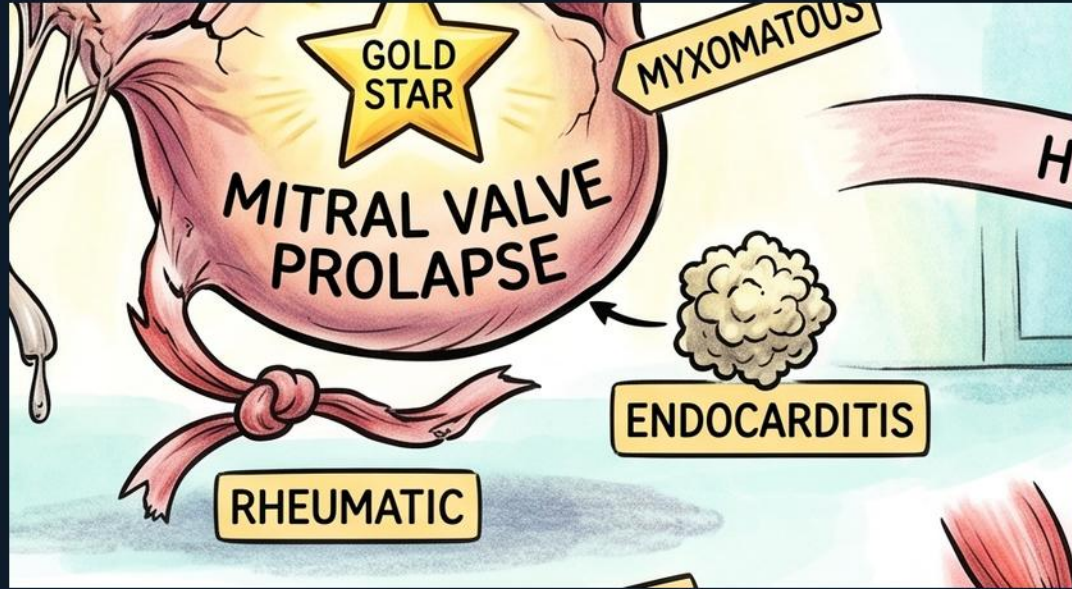
A frayed snapped cord on the far left labeled RUPTURED CHORDAE / SCARRED RIBBON.

WHAT IT ENCODES

Chordae tendineae rupture (spontaneous in myxomatous disease, or from trauma/ischemia) causes a flail leaflet and acute or chronic primary MR. It is a classic cause of sudden severe regurgitation.

THE BOARD TRAP

Overlooking flail leaflet from ruptured chordae as a cause of acute primary MR presenting with pulmonary edema.



Rheumatic primary MR

WHAT YOU SEE

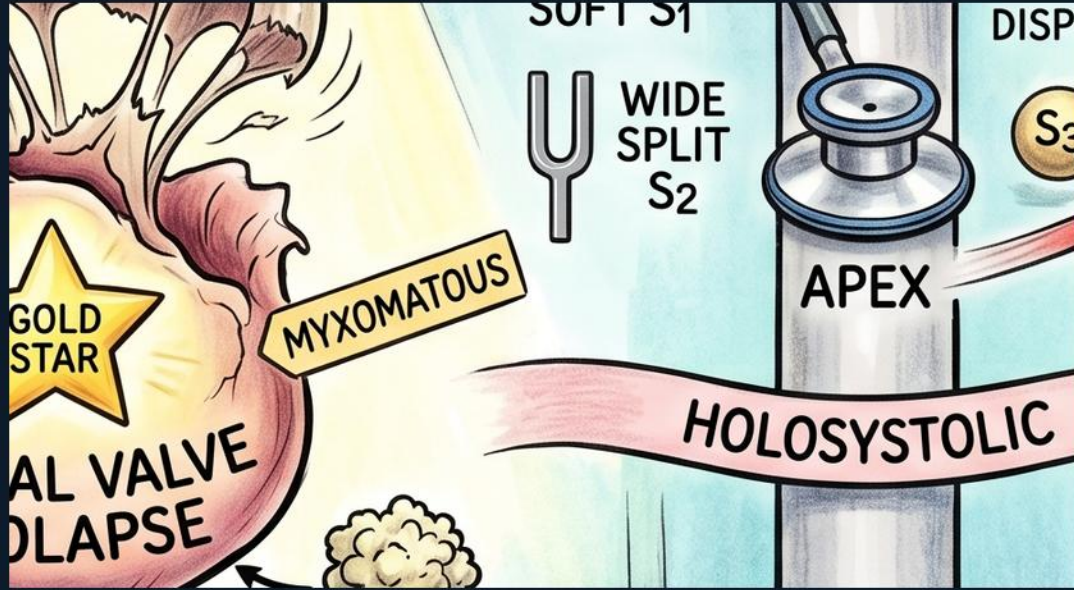
Tag labeled RHEUMATIC beneath the prolapsing valve.

WHAT IT ENCODES

Rheumatic disease can scar and retract the mitral leaflets, producing primary MR (and is still a leading cause worldwide). It is one of several organic valve etiologies in the primary group.

THE BOARD TRAP

Forgetting rheumatic disease remains a top cause of primary MR globally even where MVP dominates.



Endocarditis

WHAT YOU SEE

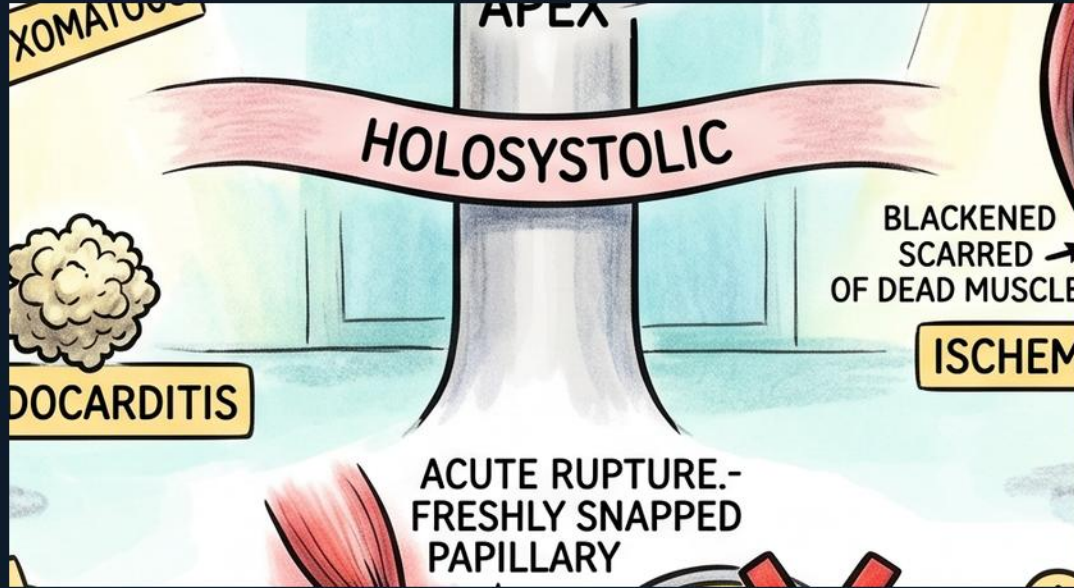
A vegetation-like cauliflower mass labeled ENDOCARDITIS.

WHAT IT ENCODES

Infective endocarditis destroys leaflet tissue or causes perforation, producing acute primary MR. New regurgitation with fever and emboli should raise this diagnosis.

THE BOARD TRAP

Missing endocarditis as a cause of new/acute primary MR in a febrile patient.



Holosystolic murmur at apex → axilla

WHAT YOU SEE

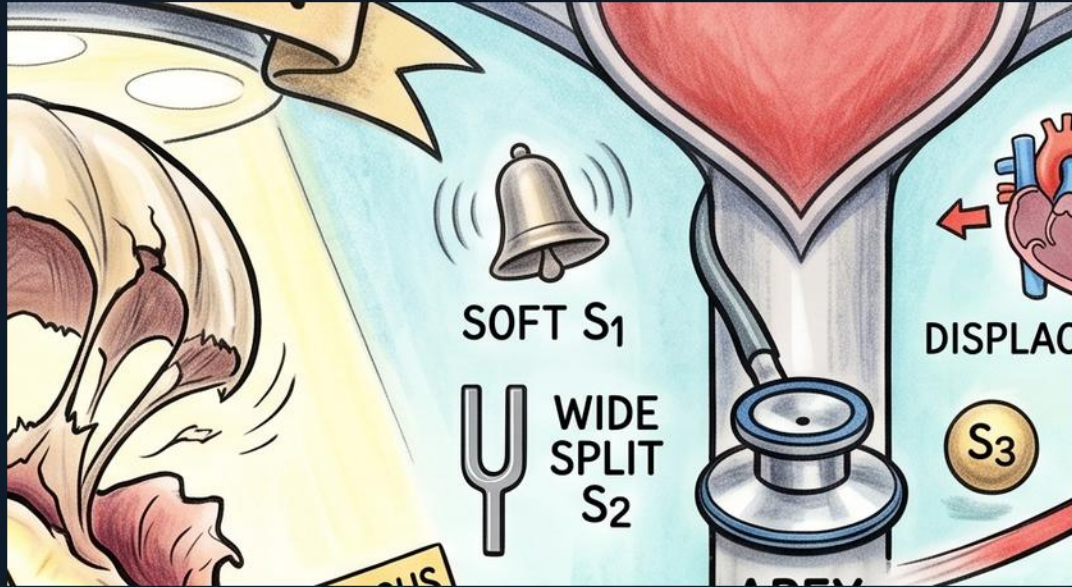
Central band labeled HOLOSYSTOLIC linking APEX to AXILLA (arrow to the right).

WHAT IT ENCODES

Chronic severe MR gives a holosystolic (pansystolic) murmur loudest at the apex radiating to the axilla. The constant pressure gradient from LV to LA throughout systole makes the murmur uniform across the whole of systole.

THE BOARD TRAP

Expecting the murmur to crescendo–decrescendo like AS — chronic MR is holosystolic, plateau in shape.



Soft / absent S1

WHAT YOU SEE

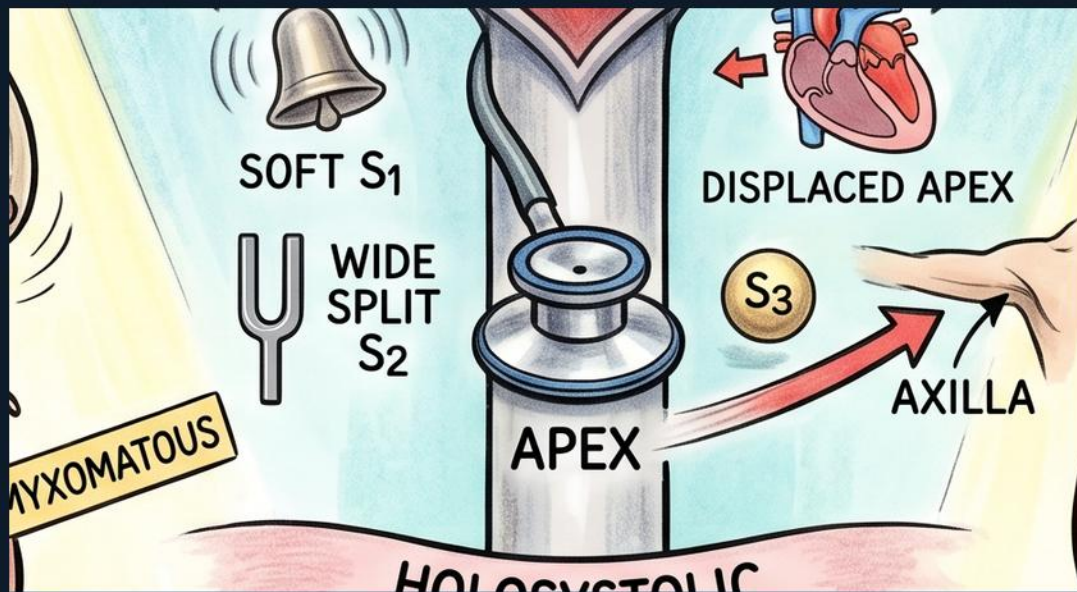
Label SOFT S1 near a bell at the top center.

WHAT IT ENCODES

With poor leaflet coaptation in chronic MR, valve closure is incomplete, so S1 is soft or absent. It is a supportive sign of significant regurgitation.

THE BOARD TRAP

Expecting a loud S1 — in MR the incompetent valve gives a SOFT/absent S1.



Widely split S2

WHAT YOU SEE

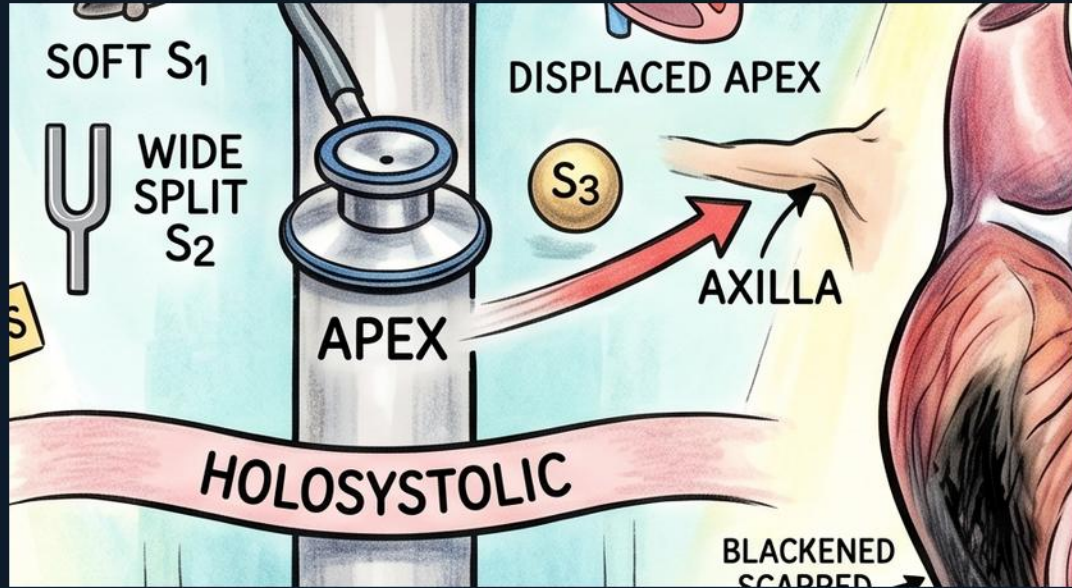
Label WIDE SPLIT S2 at the top center.

WHAT IT ENCODES

Severe MR shortens LV ejection time and brings aortic valve closure (A2) earlier, widening the A2–P2 split. It reflects rapid LV emptying into the low-pressure atrium.

THE BOARD TRAP

Reading the wide split as pulmonary disease — in MR it is from early A2 due to rapid LV unloading.



S3 gallop

WHAT YOU SEE

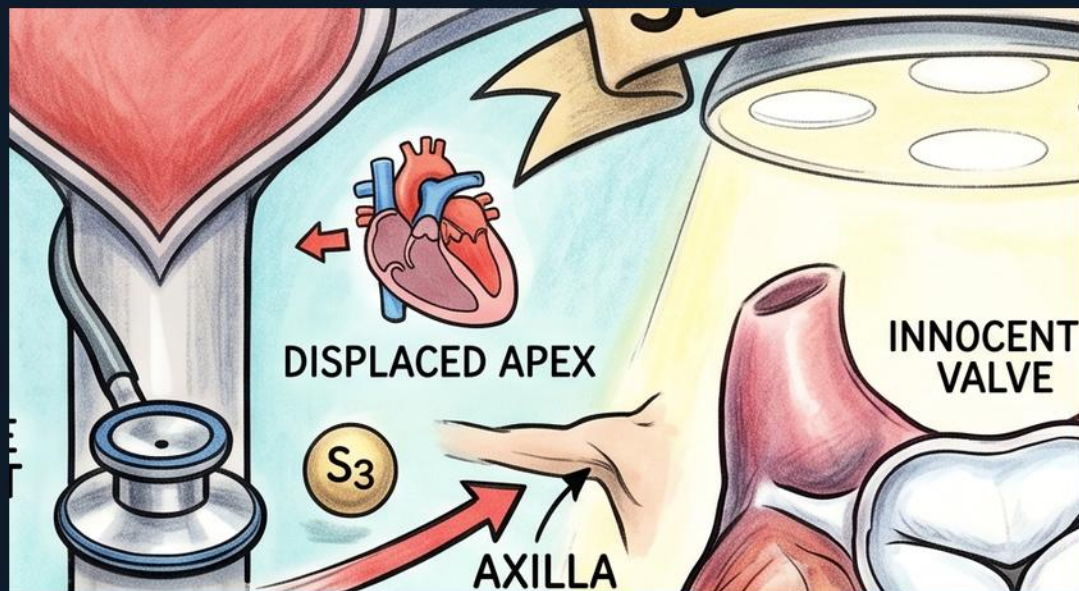
Circle labeled S₃ just right of center.

WHAT IT ENCODES

A third heart sound in chronic severe MR reflects the large regurgitant volume returning rapidly to fill the LV in early diastole. It is a marker of volume overload, not necessarily systolic failure.

THE BOARD TRAP

Equating an S₃ in MR with systolic heart failure — here it signals high regurgitant volume overload.



Displaced apex

WHAT YOU SEE

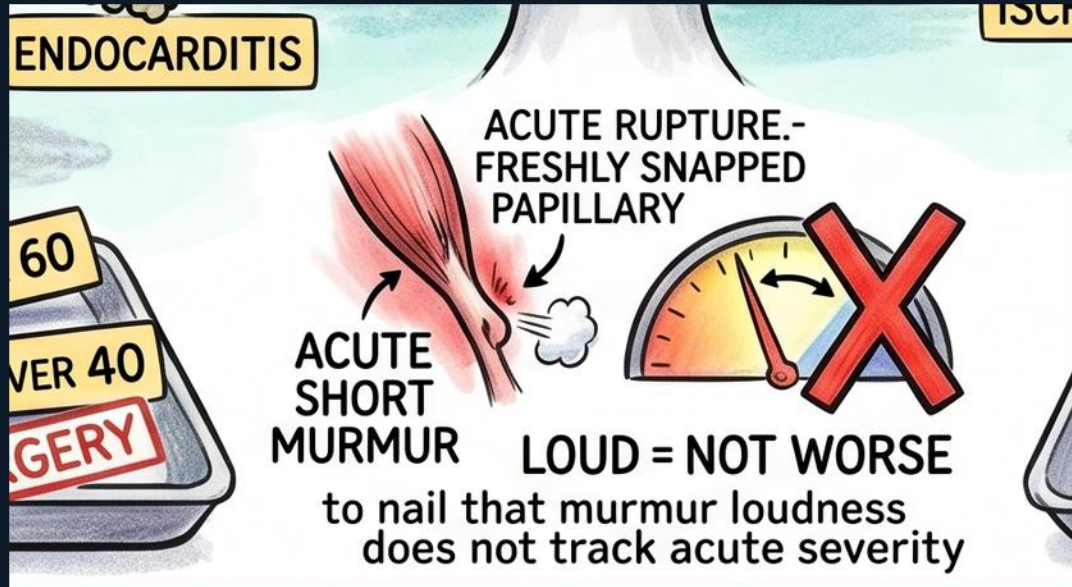
Label **DISPLACED APEX** upper center-right.

WHAT IT ENCODES

Chronic volume overload dilates the LV, displacing the apical impulse laterally and inferiorly. It signals long-standing significant MR and remodeling.

THE BOARD TRAP

Mistaking the laterally displaced, hyperdynamic apex of MR for the sustained heave of pressure overload (AS/HTN).



Acute severe MR: soft short early murmur

WHAT YOU SEE

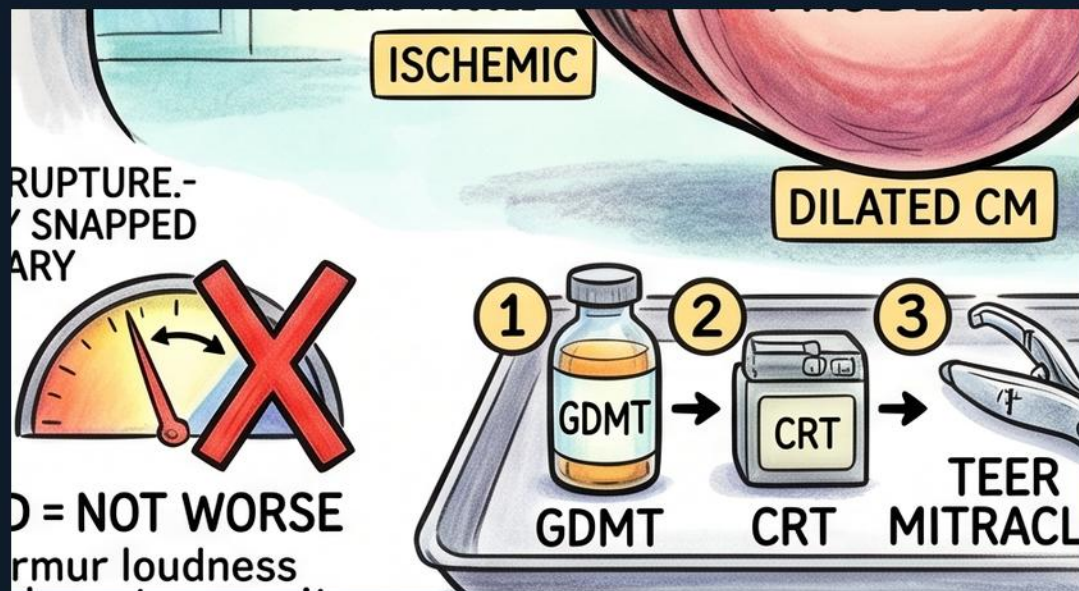
Bottom-center muscle/papillary picture labeled ACUTE RUPTURE FRESHLY SNAPPED PAPILLARY with ACUTE SHORT MURMUR.

WHAT IT ENCODES

In acute severe MR (ruptured papillary muscle or chordae), the non-compliant LA pressure rises fast, equalizing the LV-LA gradient early, so the murmur is soft, short and early — often with an S3 and flash pulmonary edema. Loudness does NOT track severity.

THE BOARD TRAP

Using murmur loudness to gauge acute MR severity — a catastrophic acute leak can be quiet/short, not loud.



Loud ≠ worse (acute)

WHAT YOU SEE

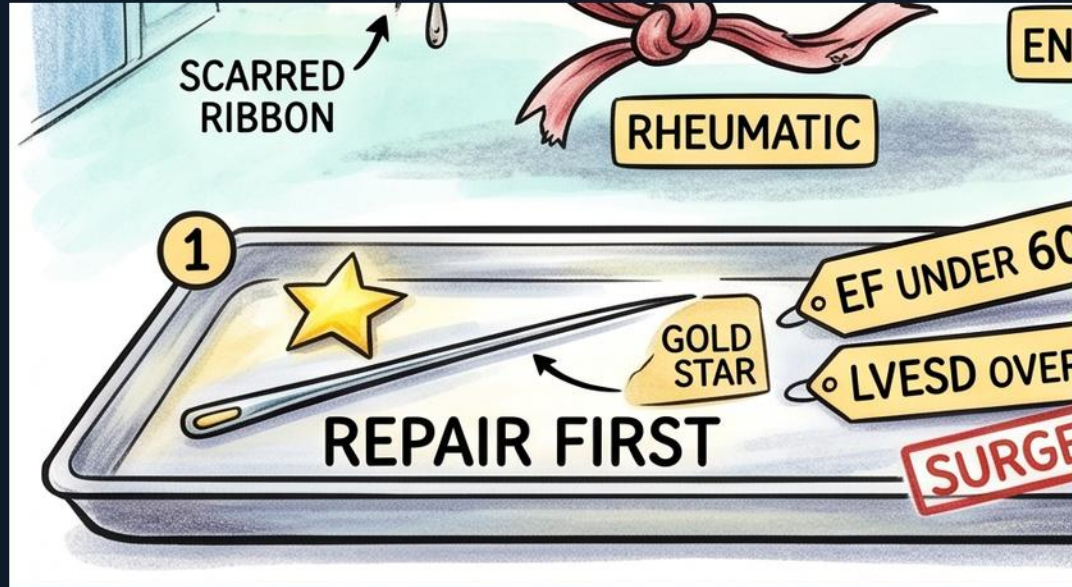
A loudness gauge with a red X, captioned LOUD = NOT WORSE / does not track acute severity.

WHAT IT ENCODES

The intensity of the regurgitant murmur correlates poorly with severity in acute MR. Severe acute regurgitation may produce only a faint murmur because LA and LV pressures rapidly equalize.

THE BOARD TRAP

Reassuring on a quiet murmur — in acute MR a soft murmur can accompany life-threatening regurgitation.



Primary severe MR → repair first

WHAT YOU SEE

Surgical tray labeled REPAIR FIRST with GOLD STAR and tag SURGERY, plus EF UNDER 60 and LVESD OVER 40.

WHAT IT ENCODES

For primary severe MR, surgery is the definitive treatment and valve REPAIR is preferred over replacement. Operate (Class I) if symptomatic, or if asymptomatic with LV dysfunction — EF 60% or less, or LV end-systolic diameter 40 mm or greater.

THE BOARD TRAP

Waiting for symptoms in primary MR — asymptomatic patients with EF $\leq 60\%$ or LVESD ≥ 40 mm already meet a Class I operative threshold.



Secondary/functional MR (ventricle problem)

WHAT YOU SEE

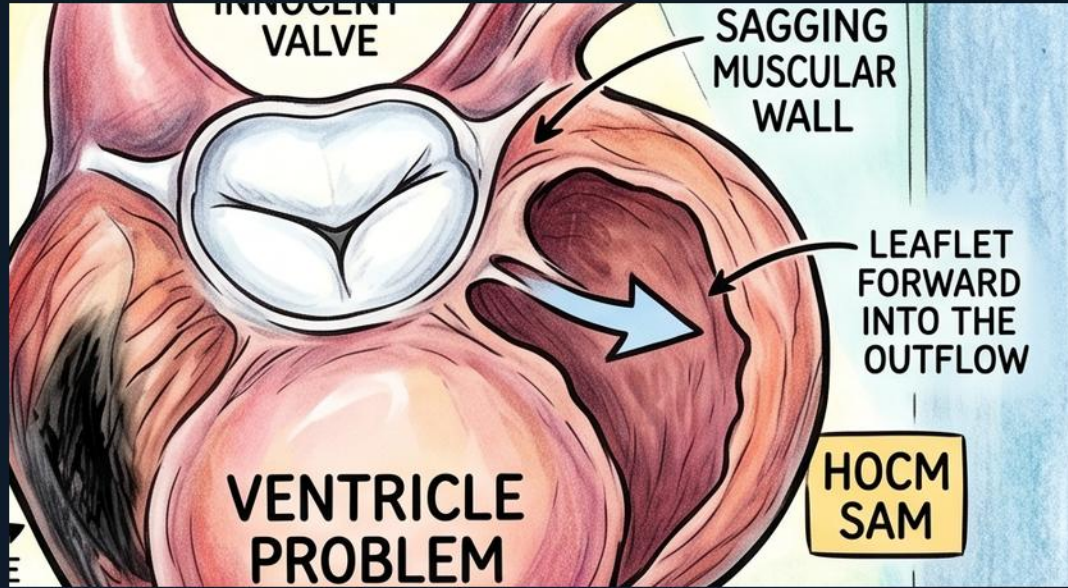
Right banner SECONDARY MR over a structurally normal valve in a dilated chamber; tag INNOCENT VALVE.

WHAT IT ENCODES

Secondary (functional) MR has a normal valve that leaks because the ventricle is diseased — annular dilation and leaflet tethering from ischemic or dilated cardiomyopathy, or HOCM with systolic anterior motion (SAM). The problem is the ventricle, not the leaflets.

THE BOARD TRAP

Operating on the valve first in functional MR — the leak is secondary to the ventricle; correct the ventricle/medical disease first.



Secondary causes: ischemic / dilated CM / HOCM-SAM

WHAT YOU SEE

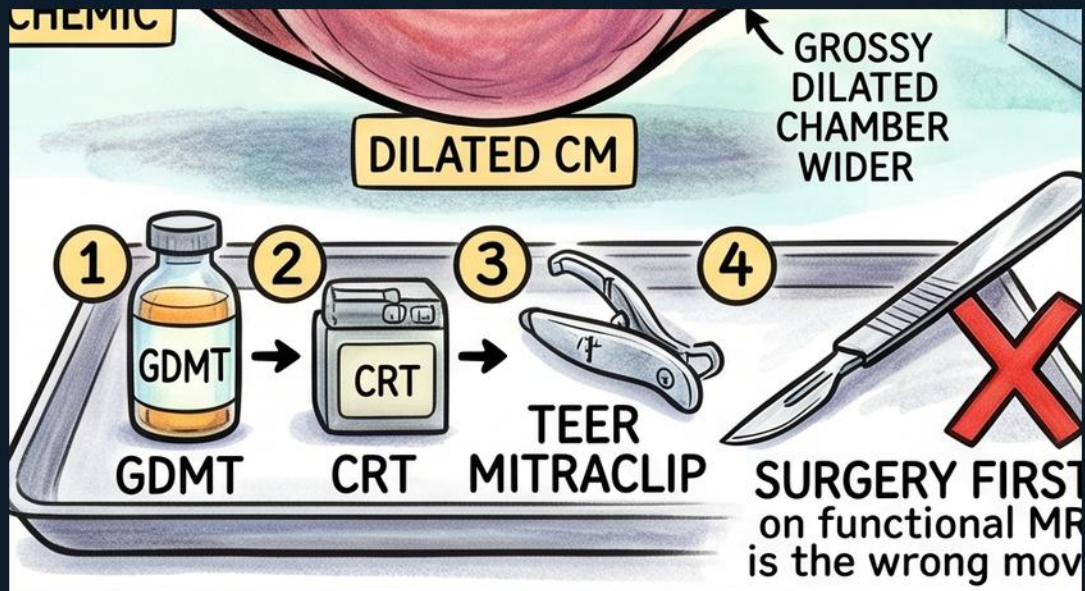
Right side tags ISCHEMIC, DILATED CM, HOCM SAM around the dilated chamber with leaflet pulled into the outflow.

WHAT IT ENCODES

Functional MR arises from ischemic cardiomyopathy, dilated cardiomyopathy, or HOCM where systolic anterior motion of the mitral leaflet causes the leak. Recognizing the underlying myopathy directs therapy at the ventricle, not the valve.

THE BOARD TRAP

In HOCM, jumping to mitral surgery — the SAM-related MR often improves with treating the obstruction, not valve replacement.



Secondary MR: GDMT → CRT → TEER

WHAT YOU SEE

Numbered steps 1 GDMT, 2 CRT, 3 TEER MITRACLIP, then 4 a scalpel with red X labeled SURGERY FIRST IS THE WRONG MOVE.

WHAT IT ENCODES

Functional MR is treated by optimizing the ventricle: guideline-directed medical therapy first, then cardiac resynchronization therapy where indicated, then transcatheter edge-to-edge repair (TEER/MitraClip) for persistent severe symptomatic MR. Primary valve surgery is not the first move.

THE BOARD TRAP

Jumping straight to valve surgery in secondary MR — the sequence is GDMT, then CRT, then TEER; surgery-first is the wrong answer.

Summary — every symbol

1. Primary MR (valve problem)

Primary (organic) MR means the valve leaflets/apparatus themselves are diseased while the ventricle is initially normal

3. Ruptured chordae

Chordae tendineae rupture (spontaneous in myxomatous disease, or from trauma/ischemia) causes a flail leaflet and acute or chronic primary MR

5. Endocarditis

Infective endocarditis destroys leaflet tissue or causes perforation, producing acute primary MR

7. Soft / absent S1

With poor leaflet coaptation in chronic MR, valve closure is incomplete, so S1 is soft or absent

9. S3 gallop

A third heart sound in chronic severe MR reflects the large regurgitant volume returning rapidly to fill the LV in early diastole

11. Acute severe MR: soft short early murmur

In acute severe MR (ruptured papillary muscle or chordae), the non-compliant LA pressure rises fast, equalizing the LV–LA gradient early, so the murmur is soft, short and early — often with an S3 and flash pulmonary edema

13. Primary severe MR → repair first

For primary severe MR, surgery is the definitive treatment and valve REPAIR is preferred over replacement

15. Secondary causes: ischemic / dilated CM / HOCM-SAM

Functional MR arises from ischemic cardiomyopathy, dilated cardiomyopathy, or HOCM where systolic anterior motion of the mitral leaflet causes the leak

2. Mitral valve prolapse / myxomatous

Myxomatous degeneration causing mitral valve prolapse is the single most common cause of primary MR in developed countries

4. Rheumatic primary MR

Rheumatic disease can scar and retract the mitral leaflets, producing primary MR (and is still a leading cause worldwide)

6. Holosystolic murmur at apex → axilla

Chronic severe MR gives a holosystolic (pansystolic) murmur loudest at the apex radiating to the axilla

8. Widely split S2

Severe MR shortens LV ejection time and brings aortic valve closure (A2) earlier, widening the A2–P2 split

10. Displaced apex

Chronic volume overload dilates the LV, displacing the apical impulse laterally and inferiorly

12. Loud ≠ worse (acute)

The intensity of the regurgitant murmur correlates poorly with severity in acute MR

14. Secondary/functional MR (ventricle problem)

Secondary (functional) MR has a normal valve that leaks because the ventricle is diseased — annular dilation and leaflet tethering from ischemic or dilated cardiomyopathy, or HOCM with systolic anterior motion (SAM)

16. Secondary MR: GDMT → CRT → TEER

Functional MR is treated by optimizing the ventricle: guideline-directed medical therapy first, then cardiac resynchronization therapy where indicated, then transcatheter edge-to-edge repair (TEER/MitraClip) for persistent severe symptomatic MR